CHEM 5260 Midterm Project
A preliminary theoretical and numerical investigation on quantum relaxation

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We study quantum relaxation in two finite-dimensional doorway-state models using Green's functions under the wide-band approximation and exact diagonalization of the full Hermitian Hamiltonian. Model A contains a single doorway state $|1\rangle$ coupled to a bath manifold, whereas Model B adds an upstream state $|0\rangle$ to produce sequential transfer $|0\rangle \to |1\rangle \to \{|k\rangle\}$. For Model A, the analysis recovers the Fermi's-golden-rule form $P_1(t) = e^{-\Gamma t}$ with $\Gamma = 2\pi|V|^2\rho$, and the numerics show that this regime is realized when $\rho^{-1} \lesssim \Gamma \ll \Delta E$; outside this window, finite-size effects, narrow bandwidth, and strong coupling produce oscillations and Poincaré recurrences. For Model B, the same reduction yields an effective damped two-level problem with overdamped, critical, and underdamped regimes separated by $V_{01} = \Gamma/4$, together with a Lorentzian detuning dependence. Random-sign bath couplings are exactly gauge-equivalent to uniform couplings, whereas Gaussian magnitude disorder creates a distribution of realization-specific decay rates, so the ensemble-averaged dynamics deviates from the single-rate FGR prediction and decays more slowly at intermediate and long times.

Keywords: quantum relaxation, doorway-state model, Fermi's golden rule, wide-band approximation, damped two-level system, Poincaré recurrence

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I. INTRODUCTION

Quantum relaxation is a fundamental phenomenon in chemical physics that underlies processes ranging from intramolecular vibrational energy redistribution and electronic energy transfer to non-radiative decay and photochemical reactions [1]. In a minimal doorway-state picture, an initially populated "bright" state is coupled to a dense manifold of background states, and population leaks out on a timescale governed by the manifold density of states and the coupling matrix elements.

The analytical theory of such processes, developed via Green's function and self-energy methods [1, 2], predicts exponential decay when the manifold is broad and dense enough to act as a quasi-continuum. The decay rate is then given by Fermi's golden rule (FGR):

$$\Gamma = 2\pi|V|^2\rho, \tag{1}$$

where ρ is the density of states of the bath manifold and V is a characteristic coupling matrix element; throughout this paper we work in units where $\hbar = 1$. This result relies on the *wide-band approximation*, which requires

$$\rho^{-1} \lesssim \Gamma \ll \Delta E. \tag{2}$$

When these inequalities fail, finite systems need not relax exponentially: recurrences, persistent oscillations, and incomplete decay can all appear. Thus irreversible Markovian relaxation is not a generic property of a finite

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bath, but a continuum-limit behavior that must be justified. A related subtlety appears when the doorway–bath couplings are disordered. Random-sign disorder can be removed by a gauge transformation on the bath basis, but Gaussian magnitude disorder generates a distribution of effective rates across realizations. The disorder average therefore takes the form $\langle e^{-\Gamma t}\rangle$, which generally differs from the mean-rate FGR form $e^{-\langle\Gamma\rangle t}$ even when each realization is nearly Markovian.

The present work addresses these questions through a combined analytical and numerical study of two finite-dimensional model Hamiltonians. Model A is the minimal doorway-state problem: a single state $|1\rangle$ coupled to a bath manifold $\{|k\rangle\}$. Model B extends this construction by adding an upstream state $|0\rangle$ coupled only to $|1\rangle$, so that the population flow proceeds sequentially: $|0\rangle \to |1\rangle \to \{|k\rangle\}$.

For Model A we recover the standard wide-band result $P_1(t) = e^{-\Gamma t}$. For Model B the same approximation reduces the problem to an effective damped two-level system for $\{|0\rangle, |1\rangle\}$ [3], with overdamped, critical, and underdamped regimes separated by $V_{01} = \Gamma/4$ and a Lorentzian dependence on the detuning $E_0 - E_1$. Exact diagonalization of the full finite Hamiltonian is then used to test these predictions as N, ΔE, V_{1k}, V_{01}, and $E_0 - E_1$ are varied, and to compare uniform, random-sign, and Gaussian couplings. Our central goals are to identify when a finite bath reproduces FGR relaxation, how the wide-band approximation breaks down, how static disorder modifies the relation between single-realization and ensemble dynamics, and how the upstream state alters the relaxation pathways.

II. MODEL HAMILTONIANS AND METHODS

A. Model Hamiltonian

1. Model A

The Hilbert space of Model A is spanned by the orthonormal basis $\{|1\rangle, |k_1\rangle, |k_2\rangle, \ldots, |k_N\rangle\}$. The full Hamiltonian is

$$\hat{H}_A = E_1 |1\rangle\langle 1| + \sum_{k=1}^{N} E_k |k\rangle\langle k| + \sum_{k=1}^{N} \big(V_{1k} |1\rangle\langle k| + V_{k1} |k\rangle\langle 1|\big), \tag{3}$$

where E_1 is the energy of the doorway state and E_k are the energies of the bath levels. All couplings are taken real so that $V_{k1} = V_{1k}$. The bath energies are distributed uniformly over an interval of width ΔE centred on E_1,

$$E_k \in \Big[E_1 - \tfrac{\Delta E}{2},\ E_1 + \tfrac{\Delta E}{2}\Big], \tag{4}$$

giving a continuum density of states $\rho = N/\Delta E$. For the uniformly spaced numerical grids used below, the exact nearest-neighbour spacing is $\Delta\omega = \Delta E/(N-1)$, so that in the large-$N$ limit one has $\rho^{-1} \approx \Delta E/N \approx \Delta\omega$. Without loss of generality we set $E_1 = 0$.

The system is initialized in the doorway state, $|\psi(0)\rangle = |1\rangle$, and the principal observable is the survival probability

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2. \tag{5}$$

We also monitor the total bath population $P_K(t) = \sum_{k=1}^{N} |\langle k|\psi(t)\rangle|^2 = 1 - P_1(t)$.

2. Model B

Model B introduces an additional state $|0\rangle$ that is coupled only to the doorway state $|1\rangle$, with no direct coupling to the bath. The Hilbert space is thus $\{|0\rangle, |1\rangle, |k_1\rangle, \ldots, |k_N\rangle\}$ and the Hamiltonian is

$$\hat{H}_B = E_0 |0\rangle\langle 0| + E_1 |1\rangle\langle 1| + V_{01} |0\rangle\langle 1| + V_{10} |1\rangle\langle 0| + \sum_{k=1}^{N} E_k |k\rangle\langle k| + \sum_{k=1}^{N} \big(V_{1k} |1\rangle\langle k| + V_{k1} |k\rangle\langle 1|\big). \tag{6}$$

The total number of levels is $N+2$ and the initial condition is $|\psi(0)\rangle = |0\rangle$. The observables are

$$P_0(t) = |\langle 0|\psi(t)\rangle|^2, \tag{7}$$

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2, \tag{8}$$

$$P_K(t) = \sum_{k=1}^{N} |\langle k|\psi(t)\rangle|^2. \tag{9}$$

The energy-level structures of both models are illustrated in Fig. 1.

3. Coupling schemes

Three coupling schemes are compared throughout.

1. *Uniform couplings*: all matrix elements equal, $V_{1k} = V$ for $k = 1, \ldots, N$.
2. *Random-sign couplings*: $V_{1k} = \pm V$ with equal probability, so that $\langle V_{1k}\rangle = 0$ and $\langle V_{1k}^2\rangle^{1/2} = V$.
3. *Gaussian couplings*: $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$, so that $\langle V_{1k}\rangle = 0$ and $\langle V_{1k}^2\rangle^{1/2} = V$.

Throughout, we denote the root-mean-square doorway–bath coupling by

$$V_{\text{rms}} \equiv \langle V_{1k}^2\rangle^{1/2},$$

so that $V_{\text{rms}} = V$ for all three schemes above. Whenever a single characteristic coupling scale is needed analytically, we write $V \equiv V_{\text{rms}}$.

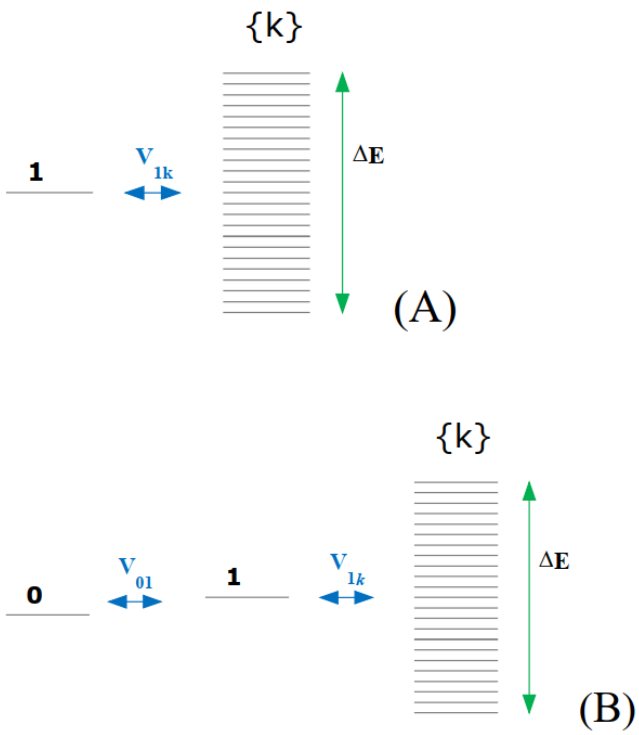

FIG. 1. Energy-level diagrams for (A) Model A, in which the doorway state $|1\rangle$ is coupled directly to the bath manifold $\{|k\rangle\}$ with matrix elements V_{1k}; and (B) Model B, in which the upstream state $|0\rangle$ is first coupled to $|1\rangle$ via V_{01}, and $|1\rangle$ subsequently couples to the manifold via V_{1k}.

B. Time-dependent Schrödinger equation and exact diagonalization

Since both Hamiltonians are represented by finite Hermitian matrices of dimension $(N+1)\times(N+1)$ (Model A) or $(N+2)\times(N+2)$ (Model B), the dynamics is governed by the time-dependent Schrödinger equation (TDSE),

$$i\hbar\frac{d}{dt}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle\,, \tag{10}$$

whose formal solution is $|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}\,|\psi(0)\rangle$. Expanding in the energy eigenbasis $\hat{H}\,|j\rangle = E_j\,|j\rangle$,

$$|\psi(t)\rangle = \sum_j e^{-iE_jt}\,|j\rangle\,\langle j|\psi(0)\rangle\,, \tag{11}$$

where $\{E_j, |j\rangle\}$ are the eigenvalues and eigenvectors of H. Numerically this spectral decomposition is obtained via the full diagonalization $H = UDU^\dagger$, which gives $e^{-iHt} = Ue^{-iDt}U^\dagger$. This approach is exact within machine precision and avoids all time-stepping errors. We work in units where $\hbar = 1$ throughout, and all parameters are reported in consistent dimensionless units. Typical system sizes range from $N = 50$ to $N = 200$ bath levels; convergence with respect to N is checked explicitly.

C. Green's function treatment and the wide-band approximation

The analytical benchmark for both models is obtained by combining the retarded Green's function with the wide-band approximation [1, 2]. For Model A this leads directly to Fermi's-golden-rule exponential decay. For Model B, after the same wide-band reduction of the bath, the remaining dynamics is that of a damped two-level system formed by the pair $\{|0\rangle\,, |1\rangle\}$.

1. *Model A: Green's function treatment under the wide-band approximation*

Equations of motion for the amplitudes. Projecting the TDSE onto the basis states of Model A and writing $c_1(t) = \langle 1|\psi(t)\rangle$ and $c_k(t) = \langle k|\psi(t)\rangle$, the coupled amplitude equations are ($\hbar = 1$)

$$i\dot{c}_1 = E_1c_1 + \sum_k V_{1k}\,c_k, \tag{12}$$

$$i\dot{c}_k = E_kc_k + V_{k1}\,c_1. \tag{13}$$

Formally integrating Eq. (13) with $c_k(0) = 0$ and substituting back into Eq. (12) yields an integro-differential equation for $c_1(t)$, whose Laplace transform is most naturally analysed via the Green's function.

Dyson equation and the self-energy. The retarded Green's operator $\hat{G}(E)$ is defined as

$$\hat{G}(E) = \lim_{\eta\to 0^+}(E - \hat{H} + i\eta)^{-1}, \tag{14}$$

and the survival amplitude is recovered by the inverse Fourier transform

$$c_1(t) = \int_{-\infty}^{\infty}\frac{dE}{2\pi i}\,e^{-iEt}\,G_{11}(E), \qquad G_{11}(E) \equiv \langle 1|\hat{G}(E)|1\rangle\,. \tag{15}$$

Projecting the resolvent identity $(E - \hat{H})\hat{G} = \hat{1}$ onto $|1\rangle$ and onto each $|k\rangle$ gives

$$G_{k1}(E) = \frac{V_{k1}}{E - E_k + i\eta}\,G_{11}(E), \tag{16}$$

and back-substituting yields the Dyson equation

$$G_{11}(E) = \frac{1}{E - E_1 - \Sigma(E)}, \tag{17}$$

where the self-energy is

$$\Sigma(E) = \sum_{k=1}^{N}\frac{|V_{1k}|^2}{E - E_k + i\eta}. \tag{18}$$

Wide-band approximation and exponential decay. In the continuum limit Eq. (18) becomes an integral over the bath density of states $\rho(\varepsilon)$:

$$\Sigma(E) \to \int_{-\infty}^{\infty} d\varepsilon\,\frac{|V(\varepsilon)|^2\rho(\varepsilon)}{E - \varepsilon + i\eta}. \tag{19}$$

Applying the Sokhotski–Plemelj identity $\lim_{\eta\to 0^+}(x+i\eta)^{-1} = \mathcal{P}(1/x) - i\pi\delta(x)$ decomposes $\Sigma = \Delta(E) - \frac{i}{2}\Gamma(E)$ with the energy-dependent decay rate

$$\Gamma(E) = 2\pi|V(E)|^2\rho(E). \tag{20}$$

The *wide-band approximation* (WBA) assumes $|V(\varepsilon)|^2\rho(\varepsilon) \approx |V|^2\rho = \text{const}$ across the resonance width, which holds when $\Gamma \ll \Delta E$ and E_1 lies well away from the band edges. The energy shift Δ then vanishes by symmetry and the self-energy reduces to a pure imaginary constant,

$$\Sigma(E) \approx -\frac{i\Gamma}{2}, \qquad \Gamma = 2\pi|V|^2\rho. \tag{21}$$

The Green's function $G_{11}(E)$ (17) becomes a Lorentzian pole,

$$G_{11}(E) \approx \frac{1}{E - E_1 + i\Gamma/2}, \tag{22}$$

of half-width $\Gamma/2$ centered at E_1. Evaluating the inverse transform (15) by closing the contour in the lower half-plane ($t > 0$) and picking up the single pole at $E = E_1 - i\Gamma/2$,

$$c_1(t) = e^{-iE_1 t}\, e^{-\Gamma t/2}, \tag{23}$$

so that the survival probability decays exponentially,

$$P_1(t) = |c_1(t)|^2 = e^{-\Gamma t}, \qquad \Gamma = 2\pi|V|^2\rho, \tag{24}$$

in agreement with Fermi's golden rule (1). The validity conditions are exactly those of Eq. (2); violations of either bound lead to deviations from exponential behavior explored numerically in the next section.

2. *Model B: damped two-level dynamics under the wide-band approximation*

For Model B the bath still couples only through the doorway state $|1\rangle$, so the same wide-band self-energy can be used after projecting onto the two-dimensional subspace $\{|0\rangle, |1\rangle\}$. The bath degrees of freedom are integrated out exactly as in Model A, yielding the effective non-Hermitian Hamiltonian

$$H_{\text{eff}}^{(B)} = \begin{pmatrix} E_0 & V_{01} \\ V_{01} & E_1 - i\Gamma/2 \end{pmatrix}, \qquad \Gamma = 2\pi|V_{1k}|^2\rho. \tag{25}$$

The crucial assumption here is the *wide-band approximation*: the bath seen by $|1\rangle$ is taken to be sufficiently broad and dense that $\Sigma(E) \approx -i\Gamma/2$ is energy independent, so the isolated doorway state would decay Markovianly as $c_1(t) \propto e^{-iE_1 t}e^{-\Gamma t/2}$. Only under this assumption can Model B be reduced to an ordinary damped two-level (Rabi) problem [3]. Equivalently, the amplitudes $c_0(t)$ and $c_1(t)$ obey

$$i\dot{c}_0 = E_0 c_0 + V_{01} c_1, \tag{26}$$

$$i\dot{c}_1 = V_{01} c_0 + \left(E_1 - i\frac{\Gamma}{2}\right) c_1. \tag{27}$$

The complex normal-mode energies are obtained from the characteristic equation of $H_{\text{eff}}^{(B)}$:

$$E_\pm = \frac{E_0 + E_1 - i\Gamma/2}{2} \pm \frac{1}{2}\sqrt{(E_0 - E_1 + i\Gamma/2)^2 + 4V_{01}^2}, \tag{28}$$

which govern the exact WBA dynamics of the amplitudes $c_0(t)$ and $c_1(t)$. At finite detuning $\Delta_{01} = E_0 - E_1$, these poles interpolate between coherent two-level oscillations and irreversible decay into the continuum. If the wide-band approximation fails, the decay of $|1\rangle$ is no longer strictly Markovian and the simple damping classification below ceases to be quantitatively valid.

The analytical treatment above relies on two linked assumptions of the wide-band approximation. First, the bath density of states and coupling strength must vary only slowly across the resonance width, so that the self-energy may be treated as effectively energy independent. Second, the bath must be sufficiently dense on the timescale of the decay that it can be treated as effectively continuous, i.e. $\rho^{-1} \lesssim \Gamma$. Violation of these assumptions through finite level spacing, narrow bandwidth, or strong coupling leads to the non-Markovian oscillations and Poincaré recurrences observed in the numerical results below.

The damping classification is most transparent on resonance, $E_0 = E_1 \equiv E_*$. Writing $c_j(t) = e^{-iE_* t}\tilde{c}_j(t)$ gives

$$i\dot{\tilde{c}}_0 = V_{01}\tilde{c}_1, \tag{29}$$

$$i\dot{\tilde{c}}_1 = V_{01}\tilde{c}_0 - i\frac{\Gamma}{2}\tilde{c}_1. \tag{30}$$

Eliminating $\tilde{c}_1$ yields

$$\ddot{\tilde{c}}_0 + \frac{\Gamma}{2}\dot{\tilde{c}}_0 + V_{01}^2\tilde{c}_0 = 0, \tag{31}$$

which is mathematically identical to the equation of a damped harmonic oscillator. The characteristic exponents are

$$\lambda_\pm = -\frac{\Gamma}{4} \pm \sqrt{\left(\frac{\Gamma}{4}\right)^2 - V_{01}^2}. \tag{32}$$

Therefore the relevant regimes are determined by comparing V_{01} with $\Gamma/4$:

1. *Overdamped regime* ($V_{01} < \Gamma/4$): defining $\Omega_{\text{od}} = \sqrt{(\Gamma/4)^2 - V_{01}^2}$,

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left[\cosh(\Omega_{\text{od}}t) + \frac{\Gamma}{4\Omega_{\text{od}}}\sinh(\Omega_{\text{od}}t)\right], \tag{33}$$

$$\tilde{c}_1(t) = -i\frac{V_{01}}{\Omega_{\text{od}}}e^{-\Gamma t/4}\sinh(\Omega_{\text{od}}t). \tag{34}$$

   Population transfer from $|0\rangle$ is monotonic, and in the strong-damping limit $V_{01} \ll \Gamma$ the slow decay rate reduces to $\gamma_{\text{eff}} \approx 2V_{01}^2/\Gamma$.

2. *Critical damping* ($V_{01} = \Gamma/4$):

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left(1 + \frac{\Gamma t}{4}\right), \qquad \tilde{c}_1(t) = -i\frac{\Gamma t}{4}e^{-\Gamma t/4}. \tag{35}$$

This is the crossover between monotonic relaxation and oscillatory population exchange.

3. *Underdamped regime* ($V_{01} > \Gamma/4$): defining $\Omega_{\mathrm{u}} = \sqrt{V_{01}^2 - (\Gamma/4)^2}$,

$$\tilde{c}_0(t) = e^{-\Gamma t/4}\left[\cos(\Omega_{\mathrm{u}} t) + \frac{\Gamma}{4\Omega_{\mathrm{u}}}\sin(\Omega_{\mathrm{u}} t)\right], \tag{36}$$

$$\tilde{c}_1(t) = -i\frac{V_{01}}{\Omega_{\mathrm{u}}}e^{-\Gamma t/4}\sin(\Omega_{\mathrm{u}} t). \tag{37}$$

The subsystem $\{|0\rangle, |1\rangle\}$ undergoes damped Rabi oscillations with oscillation frequency $2\Omega_{\mathrm{u}}$ and envelope $e^{-\Gamma t/4}$.

These three regimes provide the analytical interpretation of the Model B numerics discussed below: the bath sets the damping scale Γ, while V_{01} controls whether the upstream-to-doorway transfer is monotonic, critically damped, or oscillatory. Physically, the bath provides an effective irreversible decay channel for the doorway state $|1\rangle$ and therefore acts as a dissipative environment for the $\{|0\rangle, |1\rangle\}$ subsystem. The competition between the coherent coupling V_{01} and the dissipative rate Γ is what generates the three damping regimes.

III. RESULTS AND DISCUSSION

A. Model A: exponential decay and Fermi's golden rule

1. *Baseline dynamics*

A representative set of baseline parameters satisfying the hierarchy $\Delta E \gg V_{1k} \gg \rho^{-1}$ is

$$N = 200, \quad \Delta E = 10, \quad V_{1k} = 0.05, \quad E_1 = 0. \tag{38}$$

For these values $\rho = N/\Delta E = 20$, $\rho^{-1} = 0.05$, and the FGR decay rate is

$$\Gamma = 2\pi(0.05)^2 \times 20 \approx 0.314, \tag{39}$$

so that $\rho^{-1} \approx \Gamma/6.3 \ll \Gamma \ll \Delta E$, firmly inside the wide-band regime. Figure 2 shows $P_1(t)$ on both linear and logarithmic scales compared with the FGR benchmark $e^{-\Gamma t}$. The two curves are nearly indistinguishable throughout the observation window $t \in [0, 20]$, which spans more than six decay lengths ($5/\Gamma \approx 15.9$), confirming that the FGR prediction is quantitatively valid when the parameter hierarchy is satisfied.

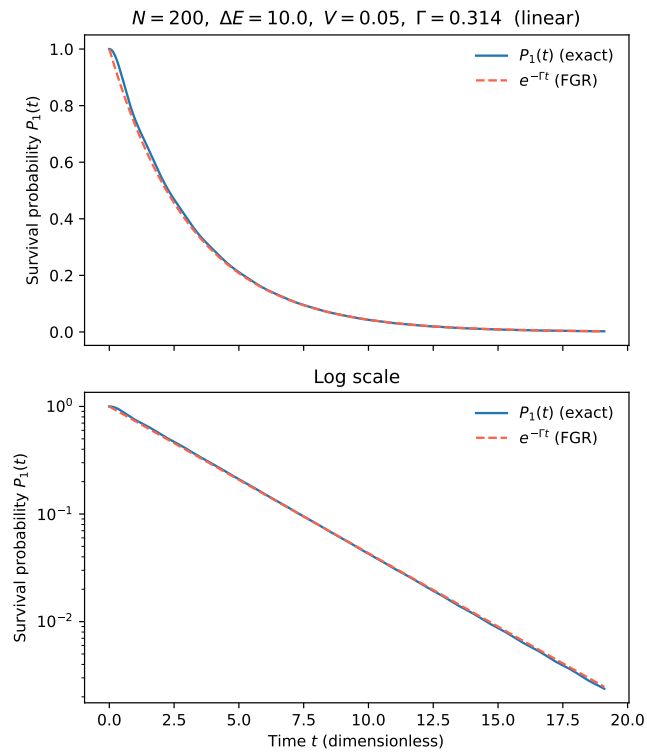

FIG. 2. Baseline dynamics of Model A with $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $E_1 = 0$ (dimensionless units, $\hbar = 1$). The density of states is $\rho = 20$, giving a FGR decay rate $\Gamma = 2\pi V_{1k}^2 \rho \approx 0.314$. Top: survival probability $P_1(t)$ (solid blue) versus the FGR prediction $e^{-\Gamma t}$ (dashed red) on a linear scale. Bottom: same data on a logarithmic scale, highlighting agreement over more than six orders of magnitude. The exact and FGR curves are virtually indistinguishable throughout the window $t \in [0, 20]$, demonstrating quantitative validity of the wide-band approximation when $\rho^{-1} \ll \Gamma \ll \Delta E$.

2. *Effect of N*

To isolate the role of the bath discreteness from a change in the decay rate, we define the *reorganization energy*

$$\Lambda \equiv \sum_{k=1}^{N} |V_{1k}|^2. \tag{40}$$

In the terminology of the present finite-level model, Λ is simply the total squared doorway–bath coupling strength. It should not be confused with the reorganization energy of shifted harmonic-oscillator models, where the same phrase refers to an energy associated with equilibrium coordinate displacements of explicit vibrational modes. For the uniform-coupling case used here, $V_{1k} = V_N$ for all k, so

$$\Lambda = N V_N^2. \tag{41}$$

We then rescale the coupling as $V_N = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ so that

$$\Lambda = NV_N^2 = N_{\rm ref}V_{\rm ref}^2 = \text{const.} \tag{42}$$

At fixed bandwidth ΔE, this also fixes the FGR rate because $\rho_N = N/\Delta E$ and therefore

$$\Gamma = 2\pi V_N^2 \rho_N = \frac{2\pi\Lambda}{\Delta E}. \tag{43}$$

Thus varying N at fixed Λ changes only the bath discreteness and the recurrence time, not the continuum-limit decay rate. For the present parameters, $\Lambda = 0.5$ and $\Gamma \approx 0.314$ for all N. The Poincaré recurrence time

$$T_{\rm rec} = \frac{2\pi}{\Delta\omega} = \frac{2\pi(N-1)}{\Delta E}, \tag{44}$$

where $\Delta\omega = \Delta E/(N-1)$ is the uniform level spacing, grows linearly with N at fixed ΔE.

Figure 3 compares $P_1(t)$ for $N = 50$, 100, and 200 (with $V_N = 0.100$, 0.071, and 0.050, respectively) over a time window extending beyond the longest $T_{\rm rec}$. For all three cases the initial decay follows the common FGR exponential $e^{-\Gamma t}$ closely, but a sharp Poincaré revival restores substantial population to $|1\rangle$ at $t \approx T_{\rm rec}$ (annotated with arrows in the figure). The revival is most prominent for small N: at $N = 50$ ($T_{\rm rec} \approx 30.8$) the peak revival amplitude exceeds 0.8, while at $N = 200$ ($T_{\rm rec} \approx 125.0$) the revival occurs far outside the experimentally relevant time window and the exponential approximation is valid over many more decay lengths. On the logarithmic scale the depth of the trough before the revival deepens with increasing N, reflecting the improved continuum-limit approximation.

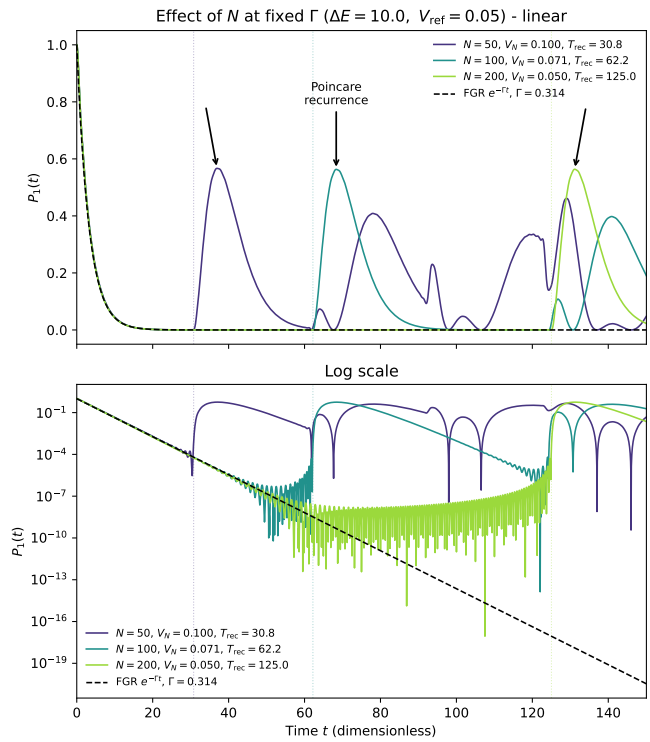

FIG. 3. Effect of the number of bath levels N on the survival probability $P_1(t)$ of Model A. The coupling is rescaled as $V_N = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ with $V_{\rm ref} = 0.05$ and $N_{\rm ref} = 200$, keeping the reorganization energy $\Lambda = NV_N^2 = 0.5$ and hence the FGR rate fixed at $\Gamma \approx 0.314$ for all curves. Parameters: $\Delta E = 10$, $E_1 = 0$. Three bath sizes are shown: $N = 50$ ($V_N = 0.100$, $T_{\rm rec} \approx 30.8$, dark blue), $N = 100$ ($V_N = 0.071$, $T_{\rm rec} \approx 62.2$, teal), and $N = 200$ ($V_N = 0.050$, $T_{\rm rec} \approx 125.0$, yellow-green). The black dashed line is the FGR prediction $e^{-\Gamma t}$. Dotted vertical lines and arrows mark the Poincaré recurrence time $T_{\rm rec}$ for each N. Top: linear scale; bottom: logarithmic scale. All curves decay exponentially at rate Γ before the first revival, demonstrating that exponential decay is an intermediate-time approximation valid only for $t \ll T_{\rm rec}$.

3. *Effect of ΔE*

At fixed $N = 200$ and $V_{1k} = 0.05$, varying ΔE simultaneously changes both the density of states $\rho = N/\Delta E$ and the ratio $\Gamma/\Delta E = 2\pi V_{1k}^2 N/\Delta E^2$, which controls how well the wide-band approximation is satisfied. Figure 4 shows $P_1(t)$ for four bandwidths $\Delta E \in \{2, 5, 10, 20\}$, corresponding to FGR rates $\Gamma \approx \{1.57, 0.63, 0.31, 0.16\}$ and ratios $\Gamma/\Delta E \approx \{0.79, 0.13, 0.03, 0.008\}$.

For the two widest bandwidths ($\Delta E = 10$ and 20) the ratio $\Gamma/\Delta E \ll 1$ and the dynamics are cleanly exponential, with $P_1(t)$ indistinguishable from the respective FGR prediction. As ΔE narrows, $\Gamma/\Delta E$ grows and the wide-band approximation begins to fail: strong oscillations are already evident at $\Delta E = 2$, the narrowest bandwidth shown, where $\Gamma/\Delta E \approx 0.79$ and the doorway state is strongly influenced by the nearby band edges. This onset of non-Markovian behavior is characterized by pronounced oscillations throughout the observation window. Conversely, larger ΔE pushes the band edges further from E_1 and deepens the Markovian limit, suppressing both oscillations and revivals.

4. *Effect of V_{1k}*

Increasing V_{1k} at fixed $N = 200$ and $\Delta E = 10$ raises Γ through $\Gamma = 2\pi V_{1k}^2\rho$, accelerating the decay while also widening the resonance relative to the bath bandwidth. Figure 5 compares three coupling strengths $V_{1k} \in \{0.05, 0.1, 0.15\}$, corresponding to $\Gamma \approx \{0.314, 1.257, 2.827\}$ and $\Gamma/\Delta E \approx \{0.03, 0.13, 0.28\}$.

For the weakest coupling ($V_{1k} = 0.05$, $\Gamma/\Delta E \approx 0.03$) the exact curve tracks the FGR exponential essentially perfectly over the entire window. At $V_{1k} = 0.1$ ($\Gamma/\Delta E \approx 0.13$) the initial decay is slightly faster than predicted by FGR and mild oscillations appear at later times, signaling incipient breakdown of the wide-band approximation. At $V_{1k} = 0.15$ ($\Gamma/\Delta E \approx 0.28$) the departure is substantial: the exact decay is markedly faster than $e^{-\Gamma t}$ at early times and persistent Rabi-like oscillations dominate the long-time dynamics, reflecting coherent coupling to the

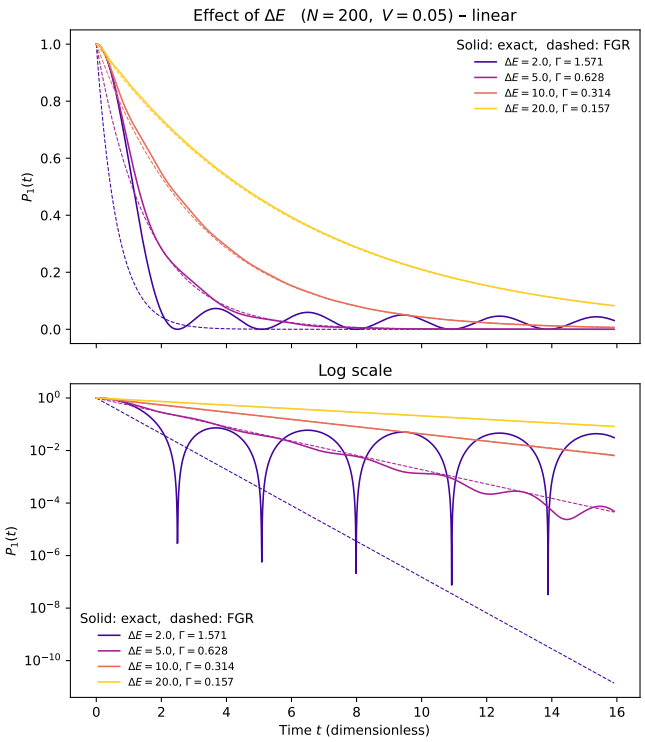

FIG. 4. Effect of the bath bandwidth ΔE on the survival probability $P_1(t)$ of Model A with $N = 200$, $V_{1k} = 0.05$, $E_1 = 0$. Four values are shown: $\Delta E = 2$ (magenta, $\Gamma \approx 1.57$), $\Delta E = 5$ (blue, $\Gamma \approx 0.63$), $\Delta E = 10$ (red, $\Gamma \approx 0.31$), $\Delta E = 20$ (yellow, $\Gamma \approx 0.16$). The black dashed line is the FGR prediction for $\Delta E = 10$. Top: linear scale; bottom: logarithmic scale. The narrowest bandwidth shown ($\Delta E = 2$, $\Gamma/\Delta E \approx 0.8$) already violates the wide-band approximation and exhibits strongly non-exponential, oscillatory dynamics. Wider bandwidths ($\Delta E \geq 10$, $\Gamma/\Delta E \leq 0.03$) satisfy the approximation and recover clean exponential decay.

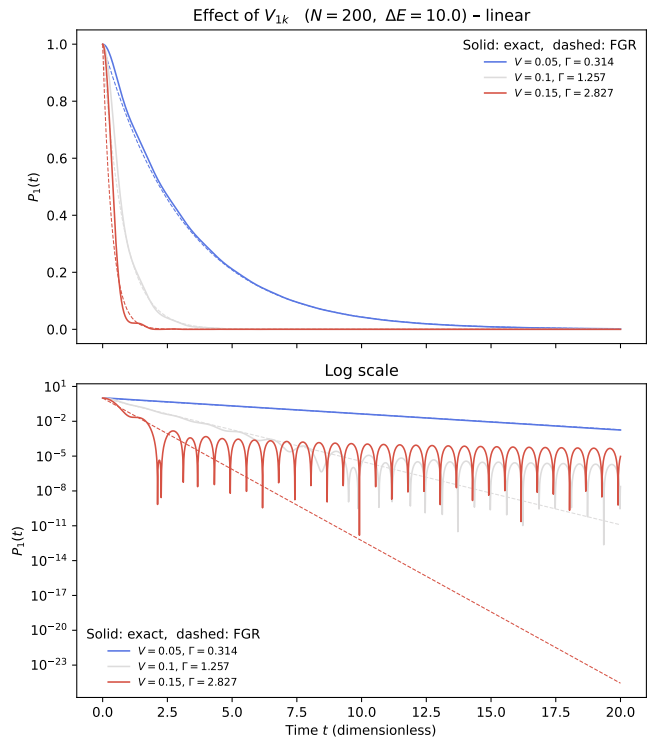

FIG. 5. Effect of the coupling strength V_{1k} on the survival probability $P_1(t)$ of Model A with $N = 200$, $\Delta E = 10$, $E_1 = 0$. Three values are shown: $V_{1k} = 0.05$ (blue, $\Gamma \approx 0.314$, $\Gamma/\Delta E \approx 0.03$), $V_{1k} = 0.10$ (silver/gray, $\Gamma \approx 1.257$, $\Gamma/\Delta E \approx 0.13$), and $V_{1k} = 0.15$ (red, $\Gamma \approx 2.827$, $\Gamma/\Delta E \approx 0.28$). Solid lines are exact TDSE results; dashed lines of matching colour are the corresponding FGR predictions $e^{-\Gamma t}$. Top: linear scale; bottom: logarithmic scale, $t \in [0, 20]$. As V_{1k} increases the resonance width Γ approaches the bandwidth ΔE, causing the exact decay to deviate progressively from the FGR exponential and eventually to exhibit sustained oscillations (crossover from Markovian-looking decay to coherent oscillatory dynamics).

collective band mode. These results confirm that the validity of FGR requires $\Gamma \ll \Delta E$, not merely $\Gamma < \Delta E$.

5. *Uniform versus random-sign couplings*

For Model A, replacing uniform couplings $V_{1k} = V$ with equal-magnitude random-sign couplings $V_{1k} = \pm V$ leaves the survival probability $P_1(t)$ *unchanged*. This is a structural consequence of the Hamiltonian topology: since each manifold state $|k\rangle$ couples only to the single doorway state $|1\rangle$, the sign of V_{1k} can be absorbed into a redefinition $|k\rangle \to -|k\rangle$, which is a valid unitary transformation that leaves all physical observables invariant. Gaussian couplings $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$ introduce seed-to-seed fluctuations in the coupling magnitudes and therefore in the realization-specific decay rates $\Gamma_r \propto \sum_k |V_{1k}|^2$. Each realization still decays approximately as $P_1^{(r)}(t) \approx e^{-\Gamma_r t}$ in the wide-band regime, but the ensemble average becomes $\langle P_1(t)\rangle = \langle e^{-\Gamma_r t}\rangle \neq e^{-\langle\Gamma\rangle t}$ whenever $\mathrm{Var}(\Gamma_r) \neq 0$. The deviation from FGR is therefore physical rather than numerical: static Gaussian disorder generates rate heterogeneity, so the averaged relaxation is a superposition of exponentials instead of a single exponential. In fact, for $t \geq 0$ Jensen's inequality gives

$$\langle e^{-\Gamma_r t}\rangle \geq e^{-t\langle\Gamma\rangle}, \tag{45}$$

because e^{-x} is strictly convex. This structure is mathematically parallel to the Jarzynski relation, where an exponential average is also controlled by fluctuations rather than by the mean alone. In the present context the exponential weighting favors rare low-Γ realizations: fast trajectories decay away quickly and cease to contribute, whereas slow trajectories dominate the ensemble average at intermediate and long times.

The inequality is saturated only if Γ_r is nonfluctuating across the ensemble, or trivially at $t = 0$. This explains why random-sign disorder, for which $|V_{1k}|^2 = V^2$ exactly, reproduces the uniform-coupling result for all times. For

Gaussian magnitude disorder the inequality is strict for any $t > 0$. In the weak-disorder or short-time regime it is nearly saturated: writing $\Gamma_r = \bar{\Gamma} + \delta\Gamma$ with $\langle \delta\Gamma \rangle = 0$, a cumulant expansion gives

$$\ln\langle e^{-\Gamma_r t}\rangle = -\bar{\Gamma}t + \frac{t^2}{2}\mathrm{Var}(\Gamma_r) - \frac{t^3}{6}\kappa_3(\Gamma_r) + \cdots , \quad (46)$$

so the leading departure from the mean-rate FGR form is controlled by $t^2\mathrm{Var}(\Gamma_r)/2$. For Gaussian couplings, $\Gamma_r \propto N^{-1}\sum_k V_{1k}^2$, so the relative variance of Γ_r scales as N^{-1}; hence the inequality becomes nearly sharp in the self-averaging large-N limit, but at long times even small residual fluctuations are amplified because the Laplace transform is dominated by the low-rate tail of the distribution.

Figure 6 shows $P_1(t)$ for all three coupling schemes ($N = 200$, $\Delta E = 10$, $V_{\mathrm{rms}} = 0.05$, averaged over 1000 disorder realisations with $\pm 1\sigma$ shaded bands). As expected, the uniform and random-sign curves are numerically identical. The Gaussian ensemble mean remains close to the FGR trend at short times, but on a logarithmic scale it bends away from a perfect straight line because it is the Laplace transform of the distribution of Γ_r. Realizations with larger $\langle V_{1k}^2\rangle$ decay faster, whereas slower realizations dominate the long-time tail, producing a decay that is slightly slower than the single-rate FGR prediction and smearing the revival structure. By contrast, random-sign disorder leaves $|V_{1k}|^2 = V^2$ exactly for every k, so all realizations share the same Γ and the ensemble average coincides with the uniform result. All three schemes satisfy $\langle\Gamma\rangle = 2\pi V_{\mathrm{rms}}^2\rho$ to within statistical uncertainty, and the residual deviation decreases as N increases and the fluctuations of Γ_r self-average away.

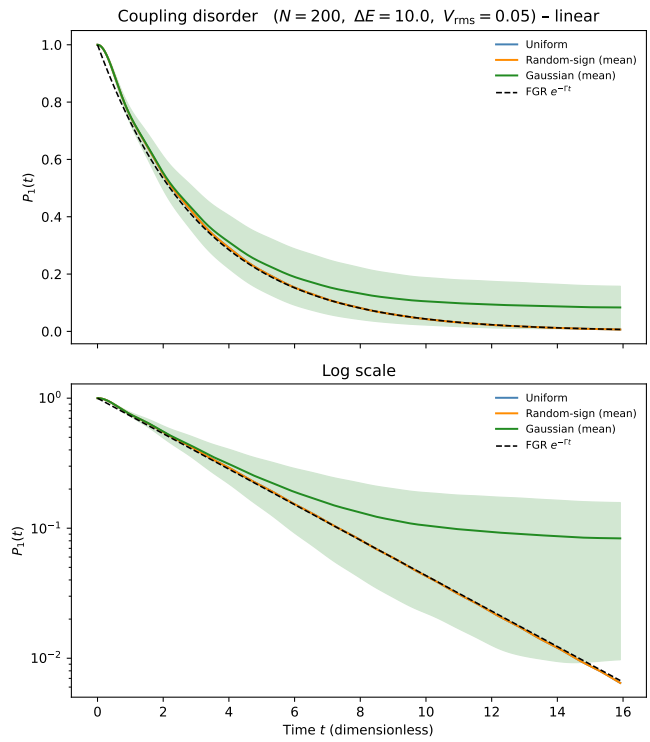

FIG. 6. Effect of coupling disorder on $P_1(t)$ in Model A ($N = 200$, $\Delta E = 10$, $V_{\mathrm{rms}} = 0.05$, $E_1 = 0$). Three coupling schemes are compared: uniform $V_{1k} = V$ (blue solid), random-sign $V_{1k} = \pm V$ (orange, mean over 1000 realisations), and Gaussian $V_{1k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, V^2)$ (green, mean over 1000 realisations). Shaded bands denote ± 1 standard deviation across realisations. The black dot-dashed line is the FGR prediction $e^{-\Gamma t}$ with $\Gamma = 2\pi V^2\rho \approx 0.314$. Top: linear scale; bottom: logarithmic scale. Uniform and random-sign results are numerically identical due to a gauge equivalence (sign redefinition of bath states). The Gaussian ensemble reproduces the same mean decay rate but exhibits broader realization-to-realization variance, especially at late times.

B. Model B: two-step relaxation dynamics

1. *Sequential population flow*

In Model B the population must flow from $|0\rangle$ to $|1\rangle$ before it can relax into the bath manifold. As derived in Sec. II C, integrating out the bath in the wide-band limit reduces the $|0\rangle$–$|1\rangle$ subspace to a damped two-level problem; on resonance, the amplitude $c_0(t)$ satisfies the same second-order equation as a damped harmonic oscillator [Eq. (31)], with the bath setting Γ and V_{01} playing the role of the restoring frequency. The critical threshold is $V_{01} = \Gamma/4 \approx 0.0785$. Figure 7 surveys five representative values of V_{01} at fixed $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$ ($\Gamma \approx 0.314$), $E_0 = E_1 = 0$, and $t_{\max} = 40$, spanning the full range from very overdamped to strongly underdamped dynamics.

Panels A and B show two overdamped cases. At $V_{01} = 0.01$ (panel A, $V_{01}/(\Gamma/4) = 0.13$), the system is deeply overdamped: the effective rate $\gamma_{\mathrm{eff}} = 2V_{01}^2/\Gamma \approx 6.4 \times 10^{-4}$ is so small that P_0 remains near unity over the entire window, P_1 is a barely visible transient, and P_K stays essentially zero. Increasing to $V_{01} = 0.04$ (panel B, $V_{01}/(\Gamma/4) = 0.51$) accelerates the decay to $\gamma_{\mathrm{eff}} \approx 0.010$; P_0 falls noticeably and a small P_1 transient develops, but the dynamics remains monotonic.

Panel C shows critical damping ($V_{01} = \Gamma/4 \approx 0.0785$), the boundary between regimes. The characteristic exponents $\lambda_\pm$ are degenerate, giving the P_0 envelope $e^{-\Gamma t/4}(1 + \Gamma t/4)$. Population reaches the bath as quickly as possible without oscillating, and the P_1 transient is larger than in either overdamped panel.

Panels D and E exhibit underdamped coherent dynamics. At $V_{01} = 0.20$ (panel D, $V_{01}/(\Gamma/4) = 2.55$), damped Rabi oscillations with frequency $2\Omega_{\mathrm{u}} = 2\sqrt{V_{01}^2 - (\Gamma/4)^2} \approx 0.37$ are clearly resolved; P_K rises in a staircase pattern that mirrors each half-cycle of population transfer. At $V_{01} = 0.50$ (panel E, $V_{01}/(\Gamma/4) = 6.37$), the oscillation frequency $2\Omega_{\mathrm{u}} \approx 0.99$ is large enough that roughly six full Rabi cycles are completed before the $e^{-\Gamma t/4}$ envelope damps the amplitude to zero, with

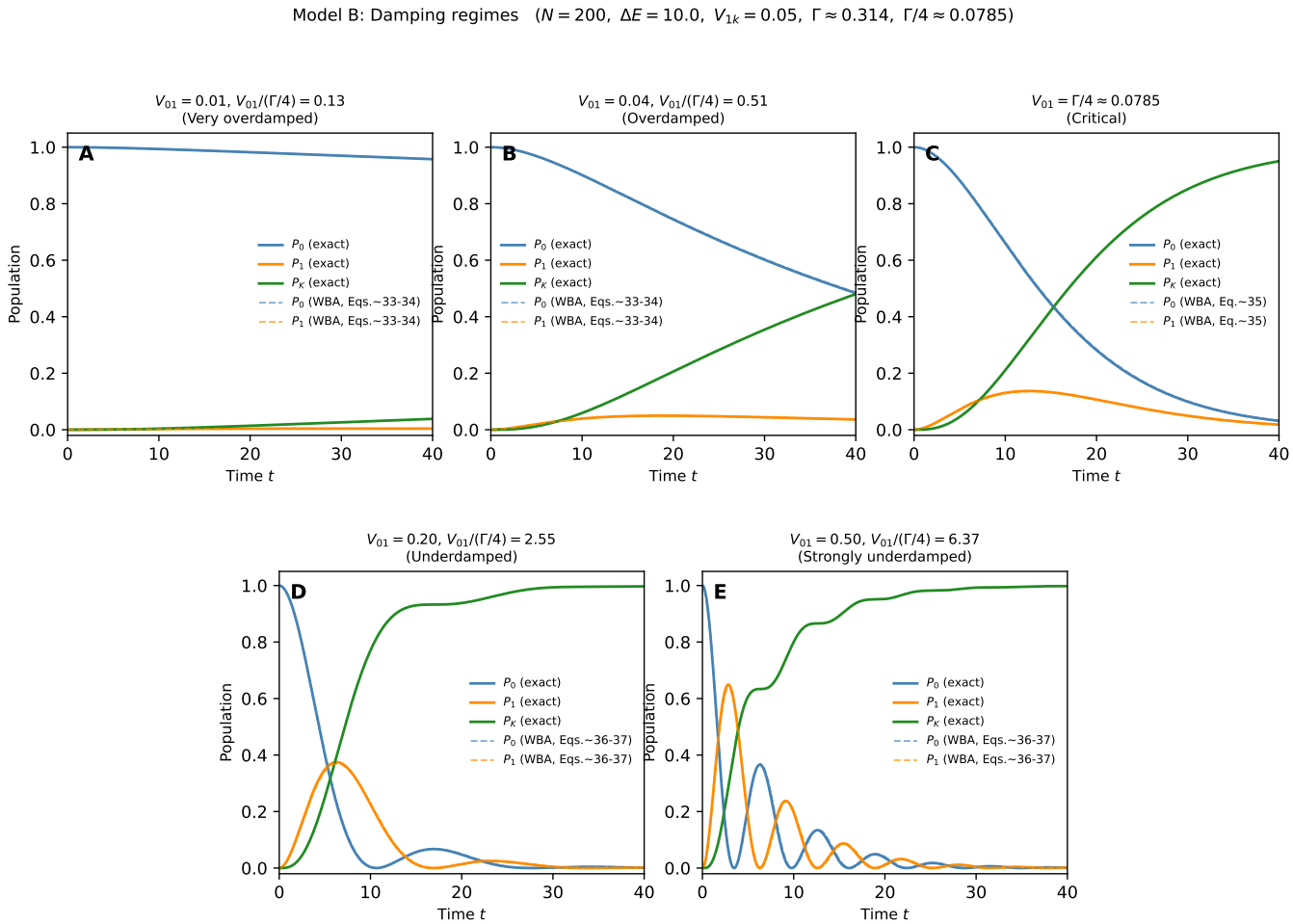

FIG. 7. **Five damping regimes of Model B**, classified by the ratio $V_{01}/(\Gamma/4)$ (Sec. II C). Parameters: $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $E_0 = E_1 = 0$, $\Gamma \approx 0.314$, $\Gamma/4 \approx 0.0785$; $t_{\max} = 40$. Solid lines: exact TDSE. Dashed lines: WBA analytical solution from Sec. II C, based on Eqs. 31 and 32. Blue: $P_0(t)$; orange: $P_1(t)$; green: $P_K(t) = 1 - P_0 - P_1$. Upper row (A–C): overdamped side of the transition. **(A)** $V_{01} = 0.01$ ($V_{01}/(\Gamma/4) = 0.13$, very overdamped): P_0 barely decays over the window; $\gamma_{\text{eff}} = 2V_{01}^2/\Gamma \approx 6.4 \times 10^{-4}$. **(B)** $V_{01} = 0.04$ ($V_{01}/(\Gamma/4) = 0.51$, overdamped): monotonic decay with $\gamma_{\text{eff}} \approx 0.010$. **(C)** $V_{01} = \Gamma/4 \approx 0.0785$ (critical): fastest non-oscillatory transfer; P_1 transient is maximal. Lower row (D–E): underdamped side. **(D)** $V_{01} = 0.20$ ($V_{01}/(\Gamma/4) = 2.55$): several damped Rabi cycles with $2\Omega_{\text{u}} \approx 0.37$. **(E)** $V_{01} = 0.50$ ($V_{01}/(\Gamma/4) = 6.37$): rapid coherent oscillations ($2\Omega_{\text{u}} \approx 0.99$, period ≈ 6.3) with full damping envelope $e^{-\Gamma t/4}$ within the window. WBA dashed lines agree with the exact TDSE in all panels.

P_K filling entirely within the window. In all five panels the WBA analytical prediction (dashed) agrees quantitatively with the exact TDSE result, validating the wide-band approximation across the full parameter range.

2. *Effect of $E_0 - E_1$*

The energy offset $\Delta_{01} = E_0 - E_1$ controls how efficiently population can flow from $|0\rangle$ to $|1\rangle$. Using the overdamped parameter $V_{01} = 0.04$ (panel B of Fig. 7), Fig. 8 scans $\Delta_{01} \in \{0, \pm 0.5, \pm 1.5, \pm 3.0\}$ with $t_{\max} = 100$, long enough to resolve the slow resonant decay $\gamma_0 = 4V_{01}^2/\Gamma \approx 0.020$.

At $\Delta_{01} = 0$ (resonance), P_0 decays at the maximum rate, P_1 builds up to its largest transient (panel A), and P_K fills up rapidly to near unity (panel B). As $|\Delta_{01}|$ increases, the effective transfer rate drops sharply: at $|\Delta_{01}| = 0.5$ it falls to $\approx 9\%$ of the resonant value, and at $|\Delta_{01}| = 3.0$ to below 0.3%. This rapid suppression is captured by the Lorentzian (panel C), whose half-width at half-maximum is the bath-induced linewidth $\Gamma/2 \approx 0.157$, far smaller than the bath bandwidth $\Delta E/2 = 5$.

A key observation is that positive and negative detunings produce *identical* $P_0(t)$ and $P_1(t)$ curves (solid and dashed lines overlap exactly in panels A and B). This follows from the energy-reflection symmetry of the model: the bath spectrum is symmetric about $E_1 = 0$, so $\Delta_{01} \to -\Delta_{01}$ is equivalent to reversing all energies, which leaves populations unchanged.

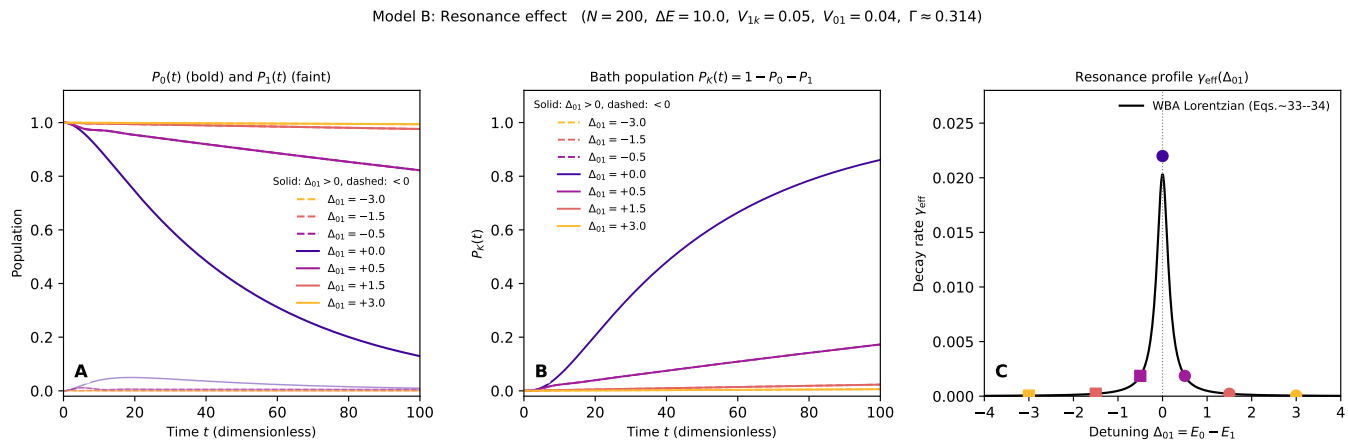

FIG. 8. **Resonance effect in Model B.** Parameters: $N = 200$, $\Delta E = 10$, $V_{1k} = 0.05$, $V_{01} = 0.04$ (overdamped, panel B of Fig. 7), $\Gamma \approx 0.314$; $t_{\max} = 100$. Detunings $\Delta_{01} \in \{0, \pm 0.5, \pm 1.5, \pm 3.0\}$; solid lines: $\Delta_{01} > 0$, dashed: $\Delta_{01} < 0$; same color for $\pm|\Delta_{01}|$. **(A)** Both $P_0(t)$ (bold) and $P_1(t)$ (faint, same color): P_0 decays fastest at resonance ($\Delta_{01} = 0$) while P_1 builds up to its largest transient; both are progressively suppressed as $|\Delta_{01}|$ grows. **(B)** Bath population $P_K(t) = 1 - P_0 - P_1$: P_K fills up rapidly at resonance and saturates near unity, while off-resonance population remains trapped in $|0\rangle$ over the window. Positive and negative detunings give identical dynamics, confirming the energy-reflection symmetry of the bath. **(C)** Extracted population decay rate $\gamma_{\rm eff}$ versus Δ_{01} (circles: $\Delta_{01} > 0$; squares: $\Delta_{01} < 0$), overlaid with the WBA Lorentzian $\gamma_{\rm eff} = V_{01}^2 \Gamma / (\Delta_{01}^2 + (\Gamma/2)^2)$ (black curve). The Lorentzian has width $\Gamma/2 \approx 0.157$ and peak $4V_{01}^2/\Gamma \approx 0.020$.

The effective decay rate plotted in panel C is extracted from the TDSE data as follows. In the overdamped regime the fast normal mode of $H_{\rm eff}$ decays on the timescale $\sim 1/\Gamma \approx 3$, leaving a single slow exponential $P_0(t) \approx A\, e^{-\gamma_{\rm eff} t}$ for $t \gg 1/\Gamma$. We therefore fit $\ln P_0(t)$ to a straight line over the second half of the simulation window ($t \in [50, 100]$), where the fast transient has fully died out and P_0 is still well above numerical noise. The slope of the linear fit gives $\gamma_{\rm eff}$ directly. This procedure is reliable provided P_0 has not yet been fully depleted, which is satisfied for all detunings shown ($P_0(100) \gtrsim 10^{-2}$ even at resonance).

Panel C demonstrates that the WBA Lorentzian prediction agrees quantitatively with the numerically extracted rates across the full scan, confirming that the Breit–Wigner line shape governs the resonance structure of the two-step decay process.

3. *Effect of N in Model B*

Figure 9 compares $N = 50$, 100, 200 with V_{1k} scaled to keep $\Gamma \approx 0.314$ and $\Lambda = N V_{1k}^2 = 0.5$ fixed, and with $V_{01} = 0.04$ (overdamped, same parameter regime as Fig. 8). The black dashed line shows the $N \to \infty$ WBA analytical result from Sec. II C: in this limit the bath is a true continuum and no Poincaré revival occurs, so P_0, P_1, P_K vary smoothly for all time. The three $P_0(t)$ curves in panel A are nearly superimposed and closely track the WBA prediction, confirming that the rate-limiting step is the $|0\rangle \to |1\rangle$ coupling, which depends only on V_{01} and is independent of the bath density of states. Panel B shows that the doorway state $P_1(t)$ carries a small transient determined by V_{01} and Γ; small revivals appear at each $T_{\rm rec}$ as population briefly returns from the bath to $|1\rangle$. Panel C confirms that P_K fills identically for all N on the initial decay, with brief dips at $T_{\rm rec}$ corresponding to the Poincaré recurrence: the discrete bath returns its population to the system before it re-enters the bath manifold. The revival amplitude is smaller than in Model A (Fig. 3) because population must traverse the extra $|1\rangle \to |0\rangle$ step to reach P_0, attenuating the recurrence signal. Larger N pushes $T_{\rm rec}$ to longer times, extending the window over which the Markovian exponential approximation is valid.

4. *Coupling disorder in Model B*

Figure 10 overlays three coupling schemes for Model B ($N = 200$, $\Delta E = 10$, $V_{1k,\rm rms} = V_{01} = 0.05$) across three panels showing $P_0(t)$ (panel A), $P_1(t)$ (panel B), and the bath population $P_K(t) = 1 - P_0 - P_1$ (panel C).

Random-sign V_{1k} is again indistinguishable from the uniform case. As in Model A, flipping the sign of an individual coupling $V_{1k} \to -V_{1k}$ is a gauge transformation on bath state $|k\rangle$ that leaves all populations unchanged, so the mean and the ensemble spread both collapse onto the uniform baseline in all three panels.

Gaussian V_{1k} introduces genuine disorder. Averaged over 1000 realisations, $P_0(t)$ acquires a moderate but well-converged sample-to-sample variance because different realisations carry different effective Γ; the variance grows over the observation window, and the ensemble-averaged dynamics deviates slightly from both the uniform/random-sign result and the corresponding wide-band single-rate expectation. $P_1(t)$ shows a larger

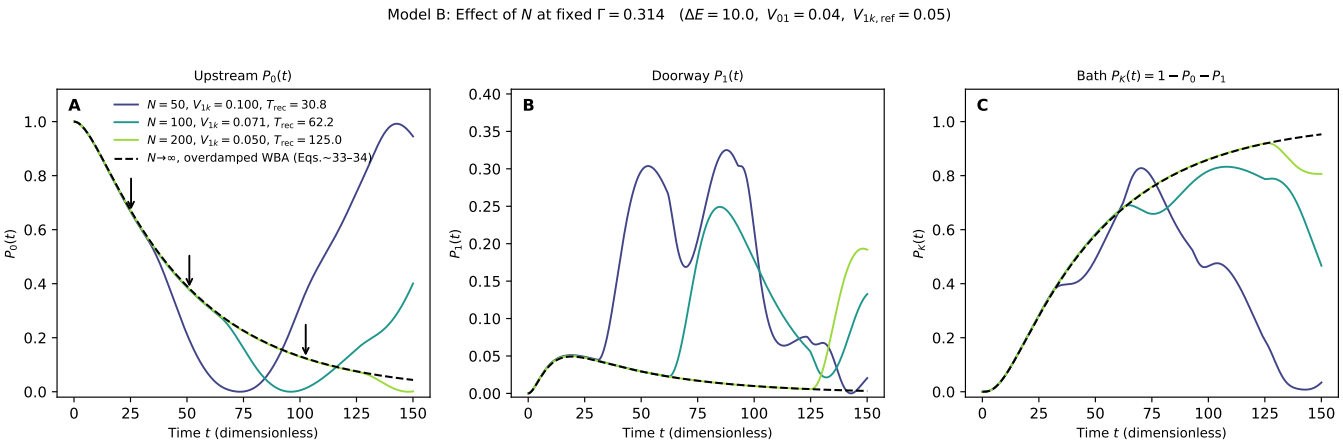

FIG. 9. **Effect of bath size N in Model B at fixed reorganization energy.** Following the design of Fig. 3, V_{1k} is scaled as $V_{1k}(N) = V_{\rm ref}\sqrt{N_{\rm ref}/N}$ with $V_{\rm ref} = 0.05$, $N_{\rm ref} = 200$ so that $\Lambda = NV_{1k}^2 = 0.5$ and $\Gamma \approx 0.314$ are held constant. The upstream coupling is fixed at $V_{01} = 0.04$ (overdamped, same as Fig. 8); $\Delta E = 10$, $E_0 = E_1 = 0$. Parameter values: $N = 50$ ($V_{1k} = 0.100$, $T_{\rm rec} = 30.8$), $N = 100$ ($V_{1k} = 0.071$, $T_{\rm rec} = 62.2$), $N = 200$ ($V_{1k} = 0.050$, $T_{\rm rec} = 125.0$). Black dashed line: $N \to \infty$ WBA analytical result from Sec. II C. Black downward arrows point to the Poincaré revival feature (local maximum in P_0, P_1; local minimum in P_K) for each N. **(A)** Upstream population $P_0(t)$: all three finite-N curves nearly coincide with the WBA limit, confirming that P_0 is rate-limited by V_{01}, not N. Revival bumps visible at each $T_{\rm rec}$ as population briefly returns from the bath. **(B)** Doorway population $P_1(t)$: revival peaks at $T_{\rm rec}$ grow in prominence as N decreases (shorter recurrence time). **(C)** Bath population $P_K(t) = 1 - P_0 - P_1$: all curves fill at the same rate; brief dips at $T_{\rm rec}$ coincide with the revivals in panels A and B.

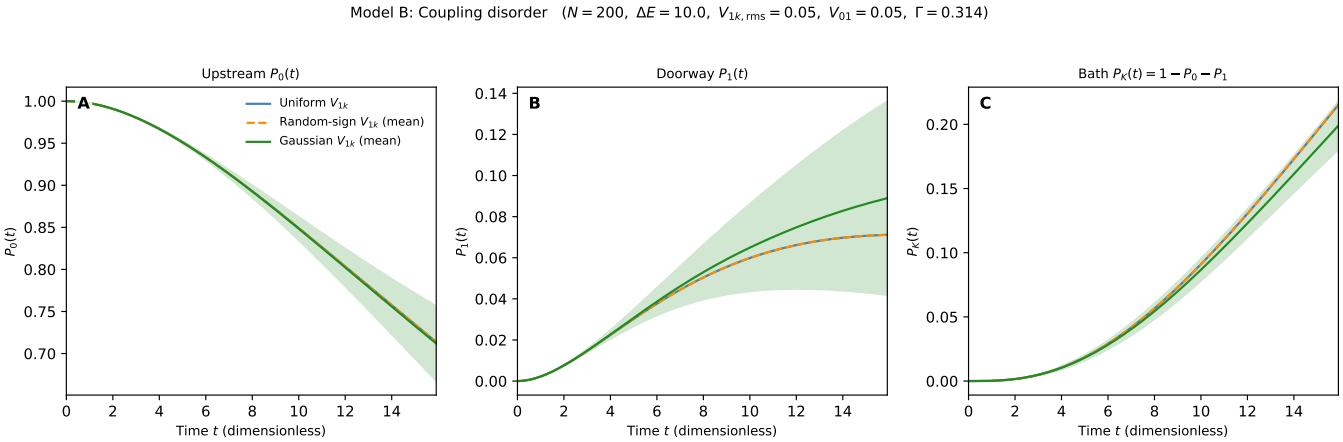

FIG. 10. **Effect of coupling disorder in Model B.** Parameters: $N = 200$, $\Delta E = 10$, $V_{1k,\rm rms} = 0.05$, $V_{01} = 0.05$, $\Gamma = 0.314$; ensemble of 1000 realisations for stochastic cases. Shaded bands show ± 1 standard deviation over realisations. Panel A: upstream population $P_0(t)$. Panel B: doorway population $P_1(t)$. Panel C: bath population $P_K(t) = 1 - P_0 - P_1$. *Blue (uniform V_{1k}):* deterministic baseline. *Orange dashed (random-sign V_{1k}):* mean coincides with the uniform curve and the band collapses to zero, confirming the gauge equivalence that extends from Model A to Model B when V_{01} is fixed. *Green (Gaussian V_{1k}):* genuine disorder shifts the ensemble mean away from the uniform/random-sign and wide-band reference dynamics, and broadens $P_1(t)$ near its transient peak; the variance is reflected symmetrically in $P_K(t)$ while leaving the slower $P_0(t)$ decay comparatively less affected.

relative variance near its transient peak: realisations with stronger bath couplings drain $|1\rangle$ faster, producing an earlier and smaller peak. The bath population $P_K(t)$ mirrors the complementary variance: when P_0 and P_1 fluctuate, the excess probability must reside in the bath, so the spread in P_K reflects the combined disorder from both upstream states.

The sign of the upstream coupling V_{01} is a global gauge freedom. A sign change $V_{01} \to -V_{01}$ is equivalent to $|0\rangle \to -|0\rangle$, which shifts the phase of the initial amplitude without affecting any population $|\langle j|\psi(t)\rangle|^2$. Randomising the sign of V_{01} therefore produces dynamics identical to the uniform case, and no disorder effect is observable.

IV. CONCLUSIONS

We have carried out a combined analytical and numerical study of quantum relaxation in two finite-dimensional doorway-state models. The principal conclusions are as follows.

1. **Model A: Green's-function theory and exponential decay.** For a single doorway state coupled to a bath manifold, the retarded Green's-function / self-energy treatment reproduces the Fermi's-golden-rule result $P_1(t) = e^{-\Gamma t}$ under the wide-band approximation. The exact TDSE numerics confirm that this Markovian exponential regime is realized when $\rho^{-1} \lesssim \Gamma \ll \Delta E$, and that the agreement improves as the bath becomes denser and broader.

2. **Finite-size effects and breakdown of the wide-band regime.** In a finite bath the exponential law is only an intermediate-time approximation. The recurrence time $T_{\rm rec} \approx 2\pi N/\Delta E$ sets the onset of quantum Poincaré recurrences, while small ΔE, small N, or large V_{1k} drive the system out of the wide-band limit and generate non-Markovian oscillations. Conversely, if V_{1k} becomes too small compared with ρ^{-1}, the manifold no longer behaves as a continuum and the FGR picture also fails.

3. **Model B: damped two-level dynamics under the wide-band approximation.** When the bath coupled to $|1\rangle$ satisfies the same wide-band conditions, Model B reduces to an effective damped two-level system for the pair $\{|0\rangle, |1\rangle\}$ only because the doorway state is assumed, under the wide-band approximation, to decay Markovianly at rate Γ. The analytical theory then predicts three regimes separated by $V_{01} = \Gamma/4$: overdamped transfer for $V_{01} < \Gamma/4$, critical damping at $V_{01} = \Gamma/4$, and damped Rabi oscillations for $V_{01} > \Gamma/4$. The exact TDSE calculations agree quantitatively with this classification over the parameter range where the WBA is valid.

4. **Resonance structure in Model B.** The energy offset $\Delta_{01} = E_0 - E_1$ acts as a detuning parameter for the upstream-to-doorway transfer. In the overdamped regime the effective decay rate follows the Lorentzian form $\gamma_{\rm eff} = V_{01}^2\Gamma/(\Delta_{01}^2 + (\Gamma/2)^2)$, so population transfer is maximal at resonance and strongly suppressed for large $|\Delta_{01}|$. Positive and negative detunings give identical dynamics due to the energy-reflection symmetry of the model.

5. **Effect of coupling disorder.** Random-sign couplings are exactly gauge-equivalent to uniform couplings and therefore leave all populations unchanged in both models. By contrast, Gaussian-distributed bath couplings introduce genuine disorder: the mean dynamics remains close to the corresponding wide-band prediction, but the ensemble average deviates systematically from a single FGR exponential because Gaussian magnitude disorder produces a distribution of realization-specific rates Γ_r. Thus $\langle e^{-\Gamma_r t}\rangle \geq e^{-\langle\Gamma\rangle t}$ by Jensen's inequality, with equality only when Γ_r does not fluctuate across realizations. The leading weak-disorder correction is $t^2{\rm Var}(\Gamma_r)/2$, while at long times the slowly decaying realizations dominate the average. The resulting sample-to-sample fluctuations broaden the envelopes and are most visible in the transient doorway population and in late-time recurrence features.

Taken together, these results provide a unified picture of doorway-state relaxation in finite systems. The wide-band approximation is the bridge from finite Hamiltonians to irreversible-looking decay, and the exact numerics show where that bridge holds and where it fails. More fundamentally, irreversible relaxation is not an intrinsic property of a finite quantum system, but an emergent continuum-limit phenomenon in which phase randomization suppresses coherent recurrences over the timescale of observation. Static Gaussian disorder adds a second, distinct mechanism for departing from the simplest FGR picture: even when each realization is locally consistent with Markovian decay, averaging over the resulting distribution of rates produces a non-exponential ensemble response in which the slow tail is enhanced.

At the same time, the present study remains intentionally minimal: the bath is represented by discrete levels, the analysis is effectively restricted to zero-temperature relaxation, and no explicit molecular vibrational coordinates are included. These simplifications clarify the origin and breakdown of the wide-band approximation, but they also limit direct chemical realism.

Future work should embed the present doorway-state picture in more microscopic open-system models. In particular, replacing the discrete phenomenological bath by an explicit oscillator environment, such as a Caldeira–Leggett model [2, 4, 5], would allow temperature effects, spectral densities, memory kernels, and vibronic structure to be treated on the same footing. Even in their present form, however, these models provide a useful minimal framework for understanding relaxation in molecular systems, nanoscale devices, and open quantum systems with structured environments.

DATA AND CODE AVAILABILITY

The data and code that support the findings of this study are available at `https://github.com/Okita0512/CHEM-5260-Midterm-Project`.

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